

CDP Reconnaissance & Flooding Attack:

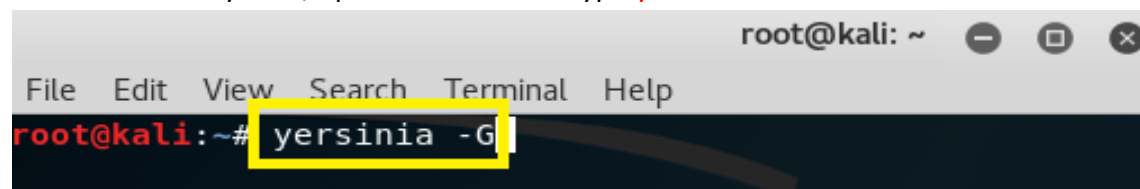
- o CDP is abbreviation, which is Stands for Cisco Discovery Protocol.
 - o Cisco Discovery Protocol (CDP) is a Layer Two-Discovery Protocol.
 - o CDP is Cisco proprietary protocol and it is identifying neighbor devices.
 - o Cisco Discovery Protocol is used for many purposes in current network.
 - o CDP identify devices and carrying VLAN information for Cisco devices.
 - o The voice VLAN in Cisco switches carried by CDP packets to IP phone.
 - o CDP message contains info about Device ID, IP, port ID, VLAN & hardware platform.
 - o CDP use multicast destination address **01-00-0c-cc-cc-cc** on each interface.
 - o Cisco Discovery Protocol CDP is enable by default in every Cisco Switch and Router.
 - o CDP advertisements are sent every 60 secs and hold time advertised is 180 seconds.
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- o Cisco Discovery Protocol is proprietary Data-Link Layer Protocol created by Cisco.
 - o It includes the Layer 3 information and its corresponding Operation System (OS).
 - o Disabled CDP to prevent an attacker getting the overall view of networking topology.
 - o CDP is used to share the information on other directly connected Cisco device.
 - o Flooding the CDP frames with fake neighbors, causing The CPU process increases.
 - o Flooding CDP frames with fake neighbors jam memory with all the neighbor entries.
 - o CDP should be disable per interface or globally to prevent an attack on Cisco devices.

```
SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

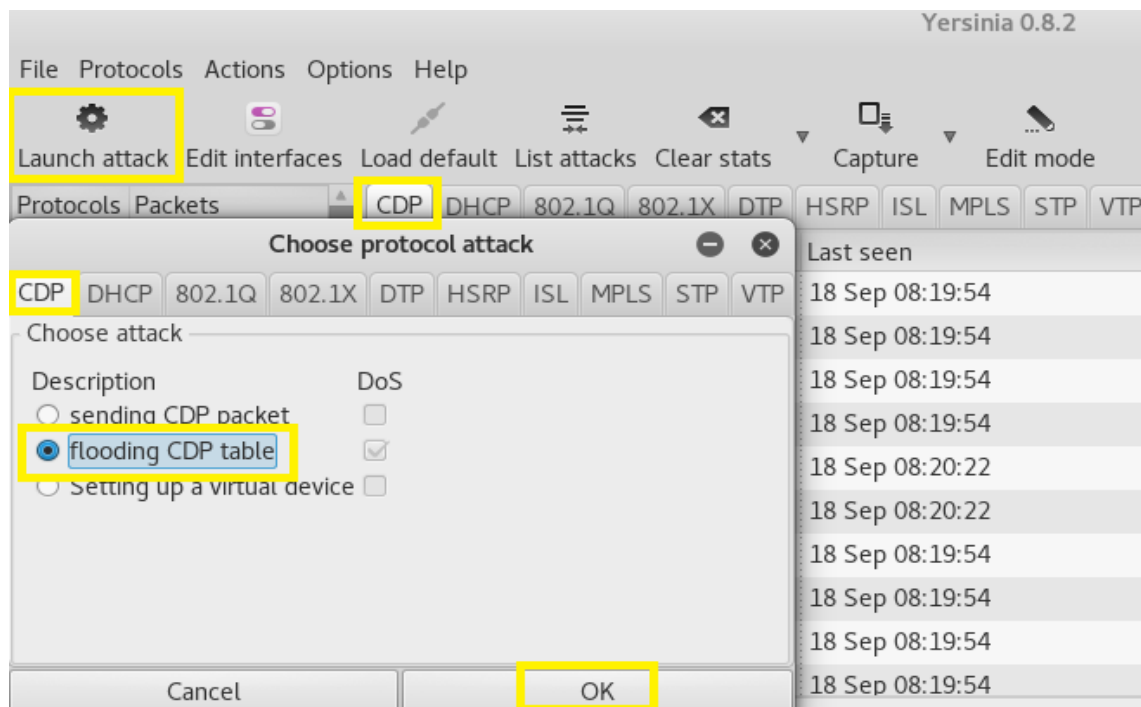
Device ID         Local Intrfce   Holdtme    Capability Platform  Port ID
SERVER            Gig 0/1         146        R S I       3640      Fas 0/0
Client            Gig 0/0         134        R S I       3640      Fas 0/0

Total cdp entries displayed : 2
```

Go to **Attacker** System, open Terminal and type **yersinia -G**.



Click on **CDP** tab, goto Main menu click on **Launch attack** in new open window select **Flooding CDP table** and press **OK** to flood the switch CDP table.

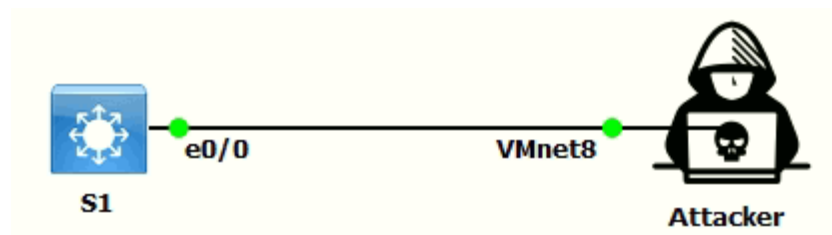


Let us check again the switch CDP neighbor table it is full of fakes CDP neighbors.

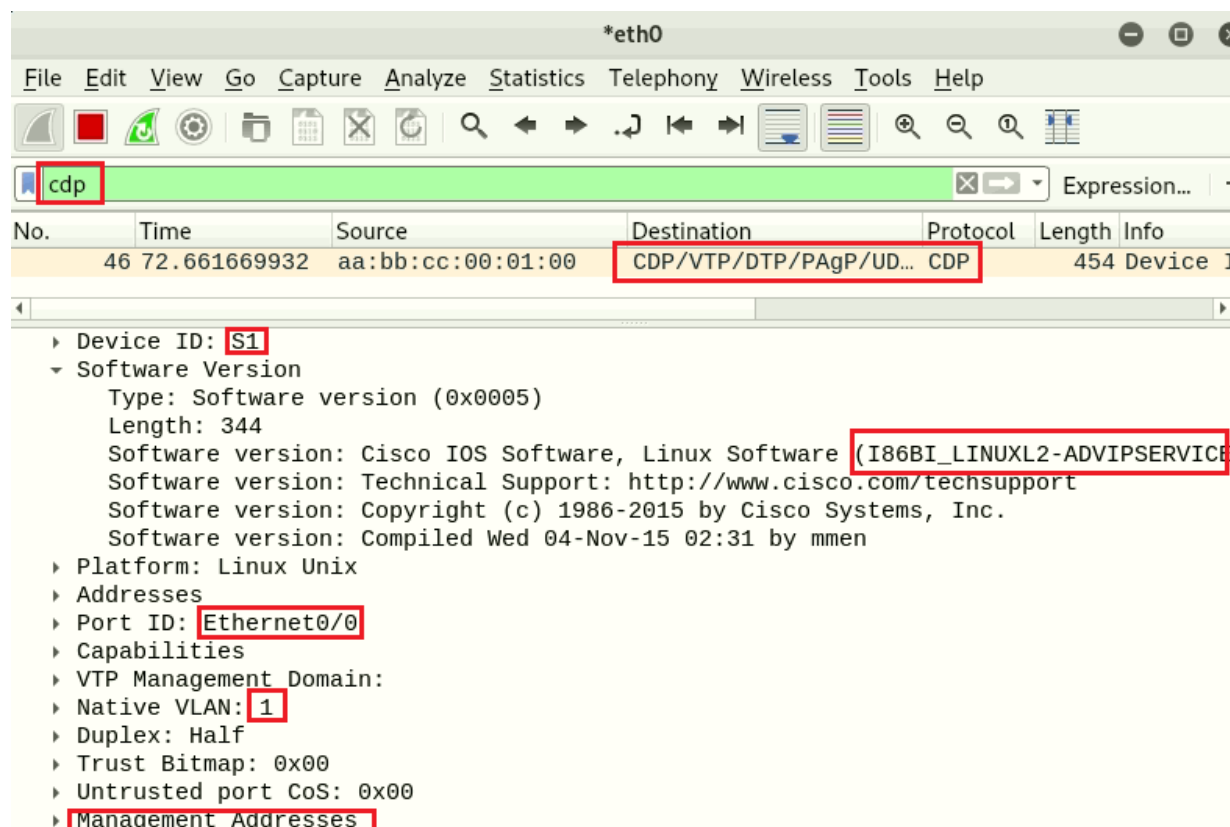
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Device ID         Local Intrfce   Holdtme    Capability   Platform    Port ID
SERVER            Gig 0/1         137        R S I        3640        Fas 0/0
R111111I          Gig 0/2         250        S H          yersinia    Eth 0
RR00000           Gig 0/2         254        R T B H      yersinia    Eth 0
0NNNNNN          Gig 0/2         253        R H I        yersinia    Eth 0
RRRRRR0           Gig 0/2         252        R T S H I    yersinia    Eth 0
0000000           Gig 0/2         252        R T B H      yersinia    Eth 0
1111111           Gig 0/2         254        T B S r      yersinia    Eth 0
JJ11111           Gig 0/2         252        T H I r      yersinia    Eth 0
NNNN222           Gig 0/2         254        R T I r      yersinia    Eth 0
JJ22222           Gig 0/2         254        R T B S      yersinia    Eth 0
2222222           Gig 0/2         252        R T H I      yersinia    Eth 0
```

CDP Reconnaissance Attack:



Just run **Wireshark** in Kali linux if want can filter only CDP packet. It will show all the details of connected Device in our case Cisco Switch hostname **S1**.



Mitigation of CDP Attack	
Commands	Description
SW1# clear cdp counters	Clear CDP counter
SW1# clear cdp table	Clear CDP table entries
SW1(config)# interface g0/2	Enter the interface mode
SW1(config-if)# no cdp enable	Disable CDP for this interface only
SW1(config)# no cdp run	Disabling CDP on the switch